```
src/
├── api/
|   └── adminAuthService.js          # Axios instance + Admin auth (login, logout, me)
|
├── components/
|   └── sidebar/
|       ├── ModeratorSidebar.jsx     # Sidebar for MODERATOR role
|       ├── SuperAdminSidebar.jsx    # Sidebar for SUPER_ADMIN role
|       ├── NavButton.jsx            # Shared nav button used by both sidebars
|       └── index.js                 # Barrel export: exports both sidebars
|
├── pages/
|   ├── auth/
|   |   └── LoginPage.jsx            # Login form (calls AdminAuthService.login)
|   ├── dashboard/
|   |   └── DashboardPage.jsx
|   ├── livestreams/
|   |   └── LivestreamsPage.jsx
|   ├── videos/
|   |   └── VideosPage.jsx
|   ├── bookings/
|   |   └── BookingsPage.jsx
|   ├── analytics/
|   |   └── AnalyticsPage.jsx
|   ├── viewers/
|   |   └── ViewersPage.jsx
|   └── superadmin/                  # SUPER_ADMIN only pages
|       ├── AdminUsersPage.jsx
|       ├── StreamConfigPage.jsx
|       ├── StreamEventsPage.jsx
|       ├── MigrationsPage.jsx
|       ├── AccessKeysPage.jsx
|       ├── SystemLogsPage.jsx
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|       ├── GlobalConfigPage.jsx
|       └── SecurityPage.jsx
|
├── layouts/
|   └── AdminLayout.jsx              # Wraps sidebar + main content area
|
├── routes/
|   ├── AppRouter.jsx                # React Router root with role-based guards
|   ├── ProtectedRoute.jsx           # Redirects to /login if not authenticated
|   └── RoleGuard.jsx                # Blocks page if role doesn't match
|
├── hooks/
|   ├── useAuth.js                   # Auth state: user, login(), logout()
|   └── useRole.js                   # isSuperAdmin(), isModerator() helpers
|
├── context/
|   └── AuthContext.jsx              # Global auth context provider
|
├── constants/
|   └── roles.js                     # AdminRole enum: SUPER_ADMIN, MODERATOR
|
├── App.jsx                          # Root component, wraps AuthContext + Router
├── main.jsx                         # Vite entry point
└── index.css                        # Tailwind directives


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 KEY FILE RESPONSIBILITIES
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  adminAuthService.js
    └── login()       → POST /api/auth/login  (stores JWT)
    └── logout()      → POST /api/auth/logout (clears storage)
    └── me()          → GET  /api/auth/me
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    └── getCachedRole()  → reads localStorage (sync, no request)
    └── isAuthenticated() → checks token exists

  AuthContext.jsx
    └── Provides: { user, login, logout, loading }
    └── On mount: calls AdminAuthService.me() to rehydrate session

  ProtectedRoute.jsx
    └── If not authenticated → redirect to /login

  RoleGuard.jsx
    └── If role !== required → redirect to /dashboard

  AdminLayout.jsx
    └── role === SUPER_ADMIN  → <SuperAdminSidebar />
    └── role === MODERATOR    → <ModeratorSidebar />
    └── Renders <Outlet /> for page content

  AppRouter.jsx
    └── /login                  → LoginPage          (public)
    └── /                       → AdminLayout         (protected)
    |   ├── /dashboard          → DashboardPage
    |   ├── /livestreams        → LivestreamsPage
    |   ├── /videos             → VideosPage
    |   ├── /bookings           → BookingsPage
    |   ├── /analytics          → AnalyticsPage
    |   ├── /viewers            → ViewersPage
    |   └── /super/*            → RoleGuard(SUPER_ADMIN)
    |       ├── /super/admins   → AdminUsersPage
    |       ├── /super/config   → StreamConfigPage
    |       ├── /super/events   → StreamEventsPage
    |       ├── /super/db       → MigrationsPage
    |       ├── /super/keys     → AccessKeysPage
    |       ├── /super/logs     → SystemLogsPage
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    |       ├── /super/global    → GlobalConfigPage
    |       └── /super/security  → SecurityPage


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 INSTALL DEPENDENCIES
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  npm install axios framer-motion lucide-react react-router-dom
  npm install -D tailwindcss postcss autoprefixer
  npx tailwindcss init -p



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 ENVIRONMENT
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  .env
  └── VITE_API_URL=http://localhost:8080/api

```